

Project Abstract Guidelines

Objective

Design and verify a complete RTL-based System-on-Chip (SoC) using the provided RISC-V processor core as the central processing unit.

Each student shall:

1. Use the provided RISC-V RTL processor.
2. Integrate at least 3 additional IP blocks.
3. Create a complete memory map.
4. Develop an application that runs on the processor.
5. Demonstrate functionality through simulation and verification.
6. Deliver RTL, testbenches, software, reports, and presentation.

Mandatory SoC Architecture

Every project must contain:

Required

- RISC-V Processor (provided by faculty)
- Instruction Memory
- Data Memory
- Bus Interconnect
- UART
- Timer
- GPIO

Additional IPs (minimum 3)

Students may:

- Design their own IP
- Download open-source IPs from GitHub/OpenCores
- Modify existing IPs

Examples: (These are examples. You can choose IPs of your choice)

- SPI Controller
- I2C Controller
- PWM Generator
- DMA Controller
- Watchdog Timer
- Matrix Multiplier
- FFT Accelerator
- AES Encryption Engine
- FIR Filter
- ADC Interface
- LCD Controller
- VGA Controller
- Seven Segment Display Controller
- Sensor Controller

Come up with

- 1) Abstract (2 pages)

Must contain:

- Problem Statement
- Proposed SoC Architecture
- Selected IPs

- 2) Architecture Document

- Block diagram
- Memory map
- Interface specifications
- Bus protocol details